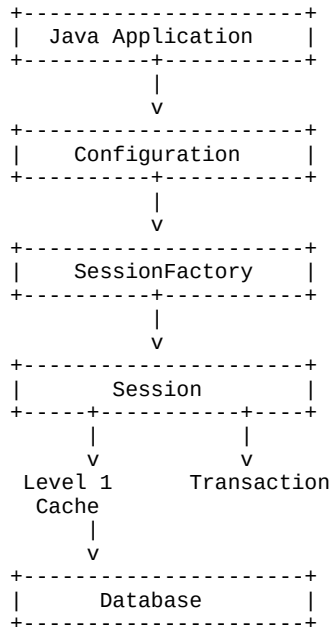


Hibernate Interview Handbook (Sample with Diagrams)

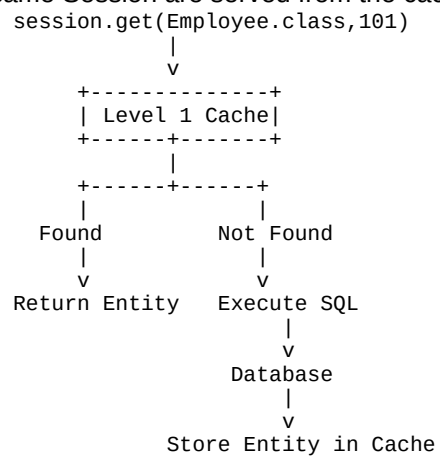
1. Hibernate Architecture

Hibernate sits between the Java application and the database. Configuration creates a SessionFactory, which creates Session objects. Each Session contains a Level 1 Cache.



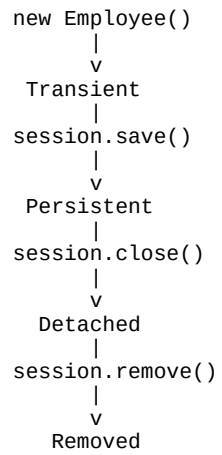
2. Level 1 Cache Flow

The first request loads the entity from the database. Subsequent requests for the same primary key in the same Session are served from the cache.



3. Entity Lifecycle

Hibernate entities move through four primary states.



4. get() vs getReference() / load()

get() loads immediately. getReference()/load() return a proxy first.

```
get()
|
+--> SQL Executes Immediately
|
+--> Record Found --> Entity
|
+--> Record Not Found --> null
```

```
getReference()/load()
|
+--> Proxy Object
|
+--> Access Non-ID Property
      |
      v
    SQL Executes
      |
+-----+-----+
|         |         |
Found    Not Found
|         |         |
Entity   Exception
|         |         |
Entity   Exception
```